N A Adarsh Pritam

adarshpritam.ml@gmail.com adarsh-crafts.github.io adarshpritam adarsh-crafts
Google Scholar

OVERVIEW

I am currently a **Research Intern at the Centre for Brain Research (CBR), Indian Institute of Science (IISc)**, where I primarily work on **machine learning methodologies for dementia research** (for early detection of cognitive decline). My research interests includes **Representation Learning, Generative Models, Computer Vision, Medical Imaging, Multimodal Learning, NeuroAI and Medical AI**.

RESEARCH EXPERIENCE

J: Journal C: Conference W: Workshop P: Preprint

Research Engineer **August 2026 – Present**
Niramai Health Analytix
- Researching foundational models for breast cancer screening using thermal imaging; computer vision, vision-language models, machine learning research and engineering.

Research Intern **May 2026 – August 2026**
Indian Institute of Science (IISc) - Centre for Brain Research (CBR)
- Conducting research on machine and statistical learning methods for early prediction of Mild Cognitive Impairment (MCI) using multimodal longitudinal cohorts, along with clinically interpretable hazard score modelling for Indian urban and rural population.
- Mentored clinicians and junior researchers in statistical and machine learning methodologies to improve interdisciplinary collaboration between clinical and AI research teams.
- **Supervisors** Dr. Thomas Gregor Issac, Dr. Palash Kumar Malo

Graduate Researcher **Aug 2025 – May 2026**
Alliance University
- Research on brain-to-image reconstruction, specifically how visual representations in the brain can be aligned with the latent spaces of modern generative and vision-language models such as latent diffusion models (LDMs) and CLIP.
- Conducted research on generative models in medical imaging, designing a systematic benchmarking framework to evaluate Generative Adversarial Networks (GANs) and diffusion models for synthetic dermoscopic image augmentation and downstream melanoma diagnosis.
- **Supervisor:** Dr. Jeba Shiney O
- **Publications:** W1 , C1 , C2

Research Intern **May 2025 – Sep 2025**
Indian Institute of Information Technology, Kottayam - Intelligent Integrated Circuits and Systems (i2CS) Research Group
- Developed a vision-language model for plant disease understanding to provide descriptive analysis from images of affected leaves.
- **Supervisor:** Dr. kala S

PUBLICATIONS

J: Journal C: Conference W: Workshop P: Preprint

W1 **Brain2VLM: Hierarchical Alignment Between Cortical Representations and Vision-Language Latent Spaces**
N A Adarsh Pritam, Jeba Shiney O, Sanyam Jain.
*European Conference on Computer Vision (**ECCV**) 2026 Workshop Proceedings & Multimodal Reasoning and Slow Thinking in the Large Model Era: Towards System 2 and Beyond Workshop @ **ECCV** 2026 (**Oral**).*

C1 **SkinGenBench: Generative Model and Preprocessing Effects for Synthetic Dermoscopic Augmentation in Melanoma Diagnosis**
N A Adarsh Pritam, Jeba Shiney O, Sanyam Jain.
IEEE International Conference on Sustainability, Innovation and Technology (IEEE - ICSIT), 2026.

C2 **American Sign Language to Fluent English Detection and Translation Using T5**
N A Adarsh Pritam, Asha Kurian.
IEEE International Conference on Emerging Technologies in Computing and Communication (IEEE - ETCC), 2025.

OPEN SOURCE PROJECTS

PyTorch Image Classification Framework
- Extended an open-source PyTorch image classification framework by integrating Grad-CAM visualization and support for modern convolutional and transformer architectures.
- **GitHub:** github.com/adarsh-crafts/pytorch-classification-extended

LLaMA from Scratch
- Implemented a LLaMA-style large language model from scratch in PyTorch, reproducing core Transformer components including grouped-query attention, rotary positional embeddings, etc.
- **GitHub:** github.com/adarsh-crafts/llama-llm-from-scratch

EDUCATION

Alliance University, Bangalore 2024 – 2026
Master of Science in Data Science **CGPA**: 8.6

Bengaluru City University, Bangalore 2021 – 2024
Bachelor of Science in Mathematics and Statistics **CGPA**: 6.5

Relevant Coursework: Machine Learning I & II, Deep Learning, ML Techniques for Image Processing, Natural Language Processing, Inferential and Descriptive Statistics, Applied Mathematics, Probability

SKILLS

Programming: Python, R, SQL
Research Areas: Representation Learning, Generative Models, Computer Vision, NeuroAI, Medical AI, Deep Learning, Machine Learning, Statistical Learning
Core Libraries: PyTorch, Torchvision, OpenCV, Transformers, Scikit-learn, Optuna, Shap, Pandas, Numpy
Tools: Git, Docker, MLflow, ZenML, AWS, Weights&Biases, Latex

CERTIFICATIONS

- AIF-C01: AWS Certified AI Practitioner - Amazon Web Services (2024)
- Fundamentals of Deep Learning - NVIDIA (2025)
- Mathematics for Machine Learning and Data Science – DeepLearning.AI (2024)

ACHIEVEMENTS

- Invited to serve on the Board of Studies (BoS) curriculum committee for the MSc Data Science program, contributing to curriculum design for subsequent (2026 onward) student cohorts.
- Invited to deliver seminars on machine learning research methodology for MSc Data Science students.